3.7 Method of variation of parameters:

Let us consider the second order linear differential equation with constants coefficient,

$$\frac{d^2y}{dx^2}+a_1\frac{dy}{dx}+a_2y=X \quad \text{.................(i)}$$

Let its complementary function be

C.F=$c_1y_1+c_2y_2$, c_1, c_2 being arbitrary constants.

Now, we introduce a determinant

$$W=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

called the Wronskian of y_1, y_2.

Then the P.I of (i) is given by

P.I=uy_1+vy_2

Where $u=-\int\frac{y_2X}{W}dx$

and $v=\int\frac{y_1X}{W}dx$

T method of solution is called the method of variation of parameters.

3.7.1 Illustrative examples:

Example 1: Solve the equation $\frac{d^2y}{dx^2}+4y=4\sec^2 2x$ by applying the method of variation of parameters.

Solution: The given equation can be written as

$$(D^2+4)y=4\sec^2 2x$$

The auxiliary equation is

$$m^2+4=0$$

$$\Rightarrow m=\pm 2i$$

$$\therefore \text{C.F}=c_1\cos 2x+c_2\sin 2x$$

Comparing this with $c_1y_1+c_2y_2$, we get

$$y_1=\cos 2x, \quad y_2=\sin 2x$$

$$\therefore y_1'=-2\sin 2x, \quad y_2'=2\cos 2x$$

$$\therefore \text{ The Wronskian W}=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$=\begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix}$$

$$=2(\cos^2 2x+\sin^2 2x)$$

$$=2$$

$$\therefore \text{ P.I}=-y_1\int\frac{y_2X}{W}dx+y_2\int\frac{y_1X}{W}dx$$

$$=-\cos 2x\int\frac{\sin 2x.4\sec^2 2x}{2}dx+\sin 2x\int\frac{\cos 2x.4\sec^2 2x}{2}dx$$

$$=-2\cos 2x\int\sec 2x\tan 2x dx+2\sin 2x\int\sec 2x dx$$

$$=-2\cos 2x\frac{\sec 2x}{2}+2\sin 2x\frac{1}{2}\log(\sec 2x+\tan 2x)$$

$$=-1+\sin 2x\log(\sec 2x+\tan 2x)$$

The general solution is,

$$y = c_1 \cos 2x + c_2 \sin 2x - 1 + \sin 2x \log(\sec 2x + \tan 2x)$$

Example 2: Solve the equation $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$ by applying the method of variation of parameters.

Solution: The given equation can be written as,

$$(D^2 - 1)y = \frac{2}{1+e^x}, \qquad \text{where } D \equiv \frac{d}{dx}$$

The auxiliary equation is,

$$m^2 - 1 = 0$$

$$\therefore m = \pm 1$$

$$\therefore \text{C.F} = c_1 e^x + c_2 e^{-x}$$

Comparing this with $c_1 y_1 + c_2 y_2$, we get

$$y_1 = e^x, \quad y_2 = e^{-x}$$

$$\therefore y_1' = e^x, \quad y_2' = -e^{-x}$$

The Wronskian $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$$= \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix}$$

$$= -e^x e^{-x} - e^x e^{-x}$$

$$= -2$$

$$\therefore \text{P.I} = -y_1 \int \frac{y_2 X}{W} dx + y_2 \int \frac{y_1 X}{W} dx$$

$$= -e^x \int \frac{e^{-x} \frac{2}{1+e^x}}{-2} dx + e^{-x} \int \frac{e^x \frac{2}{1+e^x}}{-2} dx$$

$$= e^x \int \frac{e^{-x}}{1+e^x} dx - e^{-x} \int \frac{e^x}{1+e^x} dx$$

$$= e^{x}\int\frac{dx}{e^{x}(1+e^{x})} - e^{-x}\log(1+e^{x})$$

$$= e^{x}\int\left(\frac{1}{e^{x}} - \frac{1}{1+e^{x}}\right)dx - e^{-x}\log(1+e^{x})$$

$$= e^{x}\left[\int e^{-x}dx - \int\frac{e^{-x}}{e^{-x}+1}dx\right] - e^{-x}\log(1+e^{x})$$

$$= e^{x}\left[-e^{-x} + \log(e^{-x}+1)\right] - e^{-x}\log(1+e^{x})$$

$$= -1 + e^{x}\log(e^{-x}+1) - e^{-x}\log(e^{x}+1)$$

$\therefore$ The general solution is,

$$y = c_1e^{x} + c_2e^{-x} - 1 + e^{x}\log(e^{-x}+1) - e^{-x}\log(e^{x}+1)$$

Example 3: Solve by the method of variation of parameters

$$\frac{d^2y}{dx^2} + 4y = \tan 2x$$

Solution: The given equation can be written as

$$(D^2+4)y = \tan 2x, \text{ where } D \equiv \frac{d}{dx}$$

The auxiliary equation is,

$$m^2 + 4 = 0$$

$$\Rightarrow m = \pm 2i$$

$$\therefore \text{C.F} = c_1\cos 2x + c_2\sin 2x$$

Comparing this with $c_1y_1 + c_2y_2$, we get

$$y_1 = \cos 2x, \quad y_2 = \sin 2x$$

$$\therefore y_1' = -2\sin 2x, \quad y_2' = 2\cos 2x$$

$\therefore$ The Wronskian W$=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

$$= \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix}$$

$$= 2\cos^2 2x + 2\sin^2 2x$$

$$= 2$$

$$\therefore \text{ P.I} = -y_1\int \frac{y_2 X}{W}dx + y_2\int \frac{y_1 X}{W}dx$$

$$= -\cos 2x\int \frac{\sin 2x \tan 2x}{2}dx + \sin 2x\int \frac{\cos 2x \tan 2x}{2}dx$$

$$= -\frac{1}{2}\cos 2x\int (\sec 2x - \cos 2x)dx + \frac{1}{2}\sin 2x\int \sin 2x dx$$

$$= -\frac{1}{4}\cos 2x[\log(\sec 2x + \tan 2x) - \sin 2x] - \frac{1}{4}\sin 2x \cos 2x$$

$$= -\frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$$

$\therefore$ The general solution is,

$$y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4}\cos 2x \log(\sec 2x + \tan 2x)$$

EXERCISES

Solve by the method of variation of parameters:

1. $\dfrac{d^2y}{dx^2}+y=x\sin x$

2. $\dfrac{d^2y}{dx^2}-6\dfrac{dy}{dx}+9y=\dfrac{e^{3x}}{x^2}$

3. $\dfrac{d^2y}{dx^2}-y=\left(1+e^{-x}\right)^{-2}$

4. $\dfrac{d^2y}{dx^2}+y=3x-8\cot x$